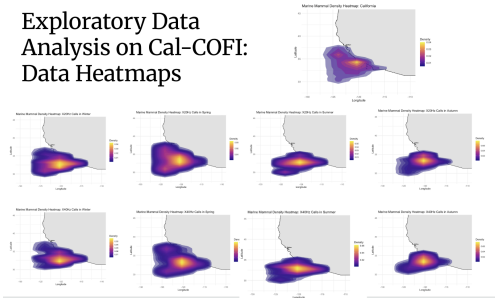
Modeling Cetacean Distribution Patterns with Spatio-Temporal GLMMS Individual Report

About Our Project. Over the course of Winter and Spring Quarter, I collaborated with five other teammates on a capstone project sponsored by Michaela Alksne, a PhD candidate at UC San Diego's Scripps Institution of Oceanography. Michaela tasked us with developing three deliverables to support her research on detecting blue and fin whale presence and understanding their use of distinct call types for key behaviors like foraging and reproduction. This research is critical for monitoring ecosystem health and informing conservation efforts within the California Current System. These deliverables include:

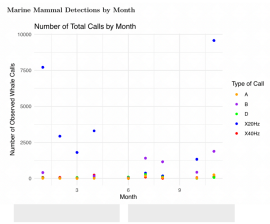
1. A spatiotemporal species distribution model (SDM) of blue and fin whale acoustic presence in the Southern California Bight.
2. Animated maps and static plots that highlight relationships between environmental predictors and whale call activity.
3. A well-documented and reproducible GitHub repository to house our data, code, and analysis.

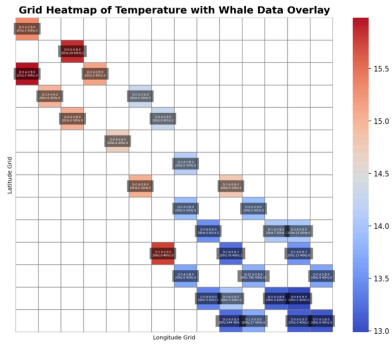
My Contribution & Deliverables.

- Spearheaded design for our final poster for the end-of-year showcase + took the initiative to reprint it through SBprinter due to previous printing issues
- Took the lead on writing the interim report, pulling together updates and insights from everyone on the team
- Served as the main organizer for the GitHub repository so it is clear, well-formatted, with a clear README, and ready to be submitted as one of our final deliverables
- Ran initial exploratory data analysis (EDA) on the CalCOFI & Acoustic datasets and built some early visualizations to help guide our modeling process
    - *Example visualizations included the following:*
    - Exploratory Data Analysis on Cal-COFI: Data Heatmaps

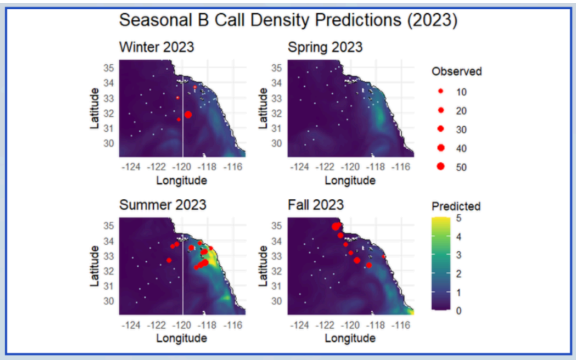


    - Exploratory Data Analysis on Cal-COFI: Calls by Month





- *Scripts edited on git: CalCOFI_EDA.rmd + Acoustic_EDA.qmd under eda folder*
- Proposed and developed a shared task and timeline [spreadsheet]() to make sure work was evenly split during our group meetings each week
- Wrote the B call modeling scripts/final deliverables:
  - Trained models for fin whale B calls to evaluate the best model based on RMSE and standard deviation and wrote about insights
  - Explored patterns between acoustic detections and environmental variables via basic visualizations
  - Conclusions:



The spatiotemporal model for predicting blue whale **B call** presence with the lowest test RMSE is :

b_scaled ~ lat + lon + temperature_depth_55 + salinity_depth_55 + magnitude_depth_55 + theta_depth_55

- **Test RMSE: 5.9131**
- **Test Standard Deviation: 6.6932**

The heatmaps on the left show that B calls are predicted to occur **most frequently in the summer** (particularly near the coast), indicating strong model prediction this season, and **least in the spring**, showing that there are low predicted activities. Fall predictions are more moderate, although there are more observed calls than predicted, so there might be a spatial mismatch.

Fig. 7: Seasonal B Call Density Predictions

  - *Scripts edited on git: B Call Final Report under docs folder, b_call folder under modeling folder*
- Drafted project individual summary + individual project presentation
  - *Scripts edited: sam_project_summary folder under individual_project_summaries folder*

Final Takeaways. While our models were unable to achieve strong predictive accuracy largely due to limitations in available data and environmental predictors, this project laid important groundwork. For future researchers hoping to expand on our work, investing more time in data collection and variable selection would likely yield stronger results.

On a personal note, this project was still incredibly rewarding, as I learned a lot about project management, team management, statistical modeling, machine learning, and scientific communication–fields that I plan to take with me as I dive into becoming a data product manager as my next stage in life.

A heartfelt thank you to Michael Alksne for being our sponsor and Dr. Baracaldo for mentoring us throughout this entire project!